Cryo-EM Project Notes

July 4, 2026

[SJ: Running AI edit: minimal cosmetic cleanup only; equations/content left unchanged except delimiter sizing, display readability, notation standardization, and theorem/proof formatting.]

"Guide to ASPIRE/scientific computing at Pton"

Math topics:

See Pere's "Introduction to Data Sciences"

3d rotations

Wavelets

L2, EMD, and WEMD

Least squares and Maximum likelihood:

1D data:

Suppose have data (x_i, y_i).

Least squares method (LS): fit a line $y = \beta x$ that gives LS error. I.e. choose $\beta = \operatorname{argmin}_\beta \sum (y_i - \beta x_i)^2$.

Explicit solution using calculus: $\frac{dE}{d\beta} = \sum -2\hat{\beta} x_i (y_i - \hat{\beta} x_i) = 0 \implies \hat{\beta} = \frac{\sum x_i y_i}{\sum x_i^2}$, the LS estimator (LSE).

Maximum likelihood method (ML): model $y = \beta x + \epsilon$, where error $\epsilon \sim N(0, \sigma^2)$. I.e. assume error $y - \beta x \sim N(0, \sigma^2)$. Then define likelihood $L(\beta)$ as probability of observing errors ϵ_i, given β, i.e.

$$\begin{aligned}
L(\beta) = P(\epsilon_i \mid \beta) = P(\epsilon_1, \ldots, \epsilon_n \mid \beta) &= \prod P(\epsilon_i \mid \beta) \\
&= \prod \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \beta x_i)^2}{2\sigma^2}\right) \\
&= \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2} \sum (y_i - \beta x_i)^2\right),
\end{aligned}$$

where we assumed iid.

Suffices to max log likelihood $\ell(\beta) = \log L(\beta) = \text{const} - \frac{1}{2\sigma^2}\sum(y_i - \beta x_i)^2$. Thus must min $\sum(y_i - \beta x_i)^2$, i.e. $\hat{\beta}_{\text{ML}} = \hat{\beta}_{\text{LS}}$.

(Link: `https://www.youtube.com/watch?v=_-Gnu498s3o`)

(Example from StatQuest: suppose have distribution of mouse weights, and want to fit normal distribution. Choose mean that max's likelihood of observing data. Then choose variance that max's likelihood of observing data.

Link: `https://www.youtube.com/watch?v=XepXtl9YKwc`)

Why are the two equivalent? As we see, both are minimizing $\sum(y_i - \beta x_i)^2$. Both use a squared distance to measure error. Different viewpoints, though. LS: fit a line, and minimize sum of squared errors. ML: assume that data lies around a line, with gaussian error (i.e. likelihood depends on squared distance).

(Link: `https://stats.stackexchange.com/questions/454742/can-someone-provide-the-intuition-as-to-why-least-squares-and-mle-lead-to-the-sa`)

Notation for conditoinal probabilities?

Higher dimensions? A row vector/matrix? See Andrew Ng notes

CTF and Fourier convolution theorem:

Convolution:

Given functions f, g of time, convolution is $(f * g)(t) = \int_{\mathbb{R}} f(\tau)g(t-\tau)d\tau$.

Intuition: Take the "weighting"/"smearing" function $g(-\tau)$. For each time t, sum the weighted values of $f(\tau)$ according to the shifted weighting function $g(t-\tau)$. Different values of $f(\tau)$ will be emphasized at different times t, based on the location of the max of $g(t-\tau)$.

Visual examples: see wikipedia. Square and triangle.

Note: if δ is the Dirac delta, $f * \delta = f$. Along with other properties, get a convolution algebra?...

Application: discrete convolution with kernel matrix can modify an image. Examples: gaussian blur, sharpen, edge detection?...

CTF: ADD from slides...

(Link: `https://en.wikipedia.org/wiki/Convolution`)

Fourier series (FS):

Given periodic function $f(x)$ on $[0,1]$ (period 1), can write as sum of exponentials (i.e. sines and cosines) with periods dividing 1. Functions $\{e^{2\pi inx}\}_{n\in\mathbb{Z}}$ form onb for IP space, so $f(x) = \sum_{n\in\mathbb{Z}} a_n e^{2\pi inx}$.

Note in term $a_n e^{2\pi i n x}$, a_n is amplitude and $e^{2\pi i n x}$ is wave with frequency n, period $\frac{1}{n}$.

Extract components using IP: $a_n = \langle f, e^{2\pi i n x}\rangle = \int_0^1 f(x)\overline{e^{2\pi i n x}}\, dx$.

Fourier transform (FT):

Suppose $f(x)$ is not periodic; can't use FS! Instead, imagine FS for period $T \to \infty$. Then $f(x) \approx \sum a_n e^{2\pi i \frac{n}{T} x}$, where each wave has period dividing T. Components become $a_n = \frac{1}{T}\int_{-T/2}^{T/2} f(x) e^{-2\pi i \frac{n}{T} x} dx$. (Divide by T because that gets isolated is integrated over length T.)

Note $a_n = \frac{1}{T}\hat{f}(\frac{n}{T})$. I.e. in the frequency domain, the frequencies are $\frac{n}{T}$, and we normalize with $\frac{1}{T}$?... The frequencies $\frac{n}{T}$ eventually become so fine that they cover all of $\mathbb{R}$.

Thus

$$f(x) \approx \sum a_n e^{2\pi i \frac{n}{T} x} = \sum \hat{f}(\frac{n}{T}) e^{2\pi i \frac{n}{T} x} \frac{1}{T} = \sum \hat{f}(\xi) e^{2\pi i \xi x} \Delta\xi \to \sum \hat{f}(\xi) e^{2\pi i \xi x} d\xi!$$

We have expressed f as the infinite sum (i.e. integral) of waves of all frequencies $\xi \in \mathbb{R}$, weighted by an infinitesimal $d\xi$; this is inverse FT.

FT gives amplitude of frequency ξ: $\hat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i \xi x} dx$.

Inverse FT writes $f(x)$ as integral of individual waves with frequencies ξ: $f(x) = \int_{\mathbb{R}} \hat{f}(\xi) e^{2\pi i \xi x} d\xi$.

In d dimensions:

Suppose $f : \mathbb{R}^2 \to \mathbb{C}$ is periodic on $[0,1]^2$. Then write as FS

$$\sum a_{m,n} e^{2\pi i m x_1} e^{2\pi i n x_2} = \sum_{(m,n)\in\mathbb{Z}^2} a_{m,n} e^{2\pi i (m,n)\cdot(x_1,x_2)} = \sum_{n\in\mathbb{Z}^2} a_n e^{2\pi i n\cdot x},$$

where the numerator has a dot product. The "frequency" is described by the vector n.

Suppose $f : \mathbb{R}^2 \to \mathbb{C}$ is not periodic. For FT, have $\hat{f}(\xi) = \int_{\mathbb{R}^2} f(x) e^{-2\pi i \xi\cdot x} dx$ to extract amplitude of "frequency" ξ. Inverse FT is $f(x) = \int_{\mathbb{R}^2} \hat{f}(\xi) e^{2\pi i \xi\cdot x} d\xi$, over all "frequencies" ξ.

(Link: `https://en.wikipedia.org/wiki/Fourier_series`)

(Link: `https://en.wikipedia.org/wiki/Fourier_transform`)

Fourier convolution theorem:

Theorem 1 (Fourier convolution theorem)**.** *Convolution in the space domain is multiplication in the Fourier domain, i.e. if $g(x), h(x)$ are functions, then*

$$\mathcal{F}(g * h) = \mathcal{F}(g) \cdot \mathcal{F}(h).$$

Vice-versa holds, too.

Proof in 1d. We have $(g * h)(x) = \int g(y)h(x-y)\,dy$, so

$$\begin{aligned}
\mathcal{F}(g * h)(\xi) &= \int (g*h)(x) e^{-2\pi i \xi x}\,dx \\
&= \int \left(\int g(y)h(x-y)\,dy \right) e^{-2\pi i \xi x}\,dx \\
&= \iint h(x-y) e^{-2\pi i \xi (x-y)} \cdot e^{-2\pi i \xi y} g(y)\,dx\,dy \\
&= \left(\int h(u) e^{-2\pi i \xi u}\,du \right) \left(\int g(y) e^{-2\pi i \xi y}\,dy \right) \\
&= \mathcal{F}(h)(\xi) \cdot \mathcal{F}(g)(\xi),
\end{aligned}$$

where $u = x - y$. □

(Link: `https://en.wikipedia.org/wiki/Convolution_theorem`)

Discrete FT (DFT, FFT):

Given sequence $x_0, \ldots, x_{n-1}$, define DFT as $X_k = \sum_0^{N-1} f(n) e^{-2\pi i \frac{k}{N} n}$ for all $0 \leqslant k \leqslant n-1$.

Motivation: think of $x_0, \ldots, x_{n-1}$ as $f(0)$ through $f(n-1)$, across an interval of period N (imagine step function). Then regular FT is $\hat{f}(k) = \int_0^N f(n) e^{-2\pi i \frac{k}{N} n} dn$ – extract frequency $\frac{k}{N}$ amplitude. Discrete, get $X_k = \sum_0^{N-1} x_n e^{-2\pi i \frac{k}{N} n}$. Note this can be rewritten as $X_k = \sum_{n=0}^{N-1} x_n \zeta^{-kn}$, where ζ is primitive N-th root of unity.

Inverse DFT: $x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{2\pi i \frac{k}{N} n}$ – frequency $\frac{k}{N}$ wave gets amplitude X_k, normalize with $\frac{1}{N}$. Note can be rewritten as $x_n = \sum_{k=0}^{N-1} X_k \zeta^{kn}$, where ζ is primitive N-th root of unity.

Matrix form: As DFT is linear map, there exists a matrix representation. Note

$$\mathcal{F} \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{N-1} \end{bmatrix} = \begin{bmatrix} \sum_{k=0}^{N-1} \zeta^{-0 \cdot k} x_k \\ \vdots \\ \sum_{k=0}^{N-1} \zeta^{-(N-1) \cdot k} x_k \end{bmatrix},$$

so the matrix form is $[\mathcal{F}] = [\zeta^{-j \cdot k}]_{j,k \in [0, N-1]}$. (A Vandermonde matrix.)

Similarly $[\mathcal{F}^{-1}] = \frac{1}{N}[\zeta^{j \cdot k}] = \frac{1}{N}[\mathcal{F}]^*$ – note the normalization. This is almost the definition of unitary – inverse is conjugate transpose (transpose doesn't matter, as symmetric)! Define $U = \frac{1}{\sqrt{N}}[\mathcal{F}]$; then U is unitary.

Thus U (i.e. $\frac{1}{\sqrt{N}}\mathcal{F}$) is unitary, and it preserves IP (Parseval)/ preserves norms (Plancherel) on $\mathbb{C}^n$.

(Link: `https://en.wikipedia.org/wiki/Discrete_Fourier_transform`)

In 2d:

To extract frequencies $\frac{u}{M}$, $\frac{v}{N}$ from FT of $M \times N$ image, use 2d DFT

$$X_{u,v} = \sum_m \sum_n x_{m,n} e^{-2\pi i \frac{u}{M} m} e^{-2\pi i \frac{v}{N} n}.$$

Inverse DFT:

$$x_{m,n} = \frac{1}{MN} \sum_u \sum_v X_{u,v} e^{2\pi i \frac{u}{M} m} e^{2\pi i \frac{v}{N} n}.$$

2d sinusoids (which form basis): look like shaded rectangular blocks, tiling space.

Can compute 2d DFT by doing 2 1d DFTs.

(Link: `https://www.youtube.com/watch?v=YYGltoYEmKo`)

Visual aspects:

Key idea: high change in certain direction in image $\iff$ big magnitudes of high frequencies along same line in FT.

FTs usually have blob of low frequencies at the center. That's the average brightness (usually high).

For real images, symmetric about $(0, 0)$ (i.e. Fourier coefficients are conjugates).

To visualize, we look at magnitude of FFT, not phase.

Rotation commutes with FT... but computationally, not quite. On computers, FT is applied to a grid made of the input image, and that causes strange "edge effects". If one side of the edge of the image has a jump in color compared to the other side, high frequencies will show up on the $u = 0$ and $v = 0$ axes of the FT.

Reduce edge effects by "windowing": crop the image into a circle that fades to the edges.

Ex of logic: if smooth an image in the horizontal direction, there will be less drastic color jumps, i.e. less high frequency in horizontal direction. On FT, high frequencies are attenuated.

Note: translating an object doesn't change FT magnitude plot. Repeating an image reinforces the same FT.

FFT: ADD...

(Link: `https://www.youtube.com/watch?v=YYGltoYEmKo`)

(Link: `https://www.cs.unm.edu/~brayer/vision/fourier.html`)

Projection slice theorem

Theorem 2 (Projection slice theorem)**.** *In 2d, projecting onto a line ℓ and then taking the 1d FT gives the same result as taking the 2d FT and then slicing through the line parallel to ℓ. Formally,*

$$F_1 P_1 = S_1 F_2.$$

The analogous 3d statement holds for projecting onto a plane and slicing the 3d FT through the parallel plane.

The 3d case is used in Computerized Tomography (CT): given many projection images from known directions, taking 2d FTs provides slices of 3d FT of density. Piece together slices to get full 3d FT, and invert to get density of organ!

Proof in 2d. WLOG show for $\ell = x$ axis given density $f(x, y)$. We use (u, v) as coordinates of frequency domain.

Let $p(x) = P_1 f(x, y) = \int_{\mathbb{R}} f(x, y)\, dy$. Then

$$\begin{aligned}(F_1 P_1(f))(u) = (F_1(p))(u) &= \int_{\mathbb{R}} p(x) e^{-2\pi i u x}\, dx \\ &= \iint f(x, y) e^{-2\pi i u x}\, dx\, dy.\end{aligned}$$

Note

$$(F_2(f))(u, v) = \iint f(x, y) e^{-2\pi i u x} e^{-2\pi i v y}\, dx\, dy.$$

Since we are slicing along the x axis, $S_1(g(u, v)) = g(u, 0)$, hence

$$\begin{aligned}(S_1 F_2(f))(u) &= \iint f(x, y) e^{-2\pi i u x} e^{-2\pi i 0 y}\, dx\, dy \\ &= \iint f(x, y) e^{-2\pi i u x}\, dx\, dy.\end{aligned}$$

Thus $F_1 P_1 = S_1 F_2$, as desired. $\square$

(Why WLOG? Translation doesn't affect P_1 or F_2. Rotation commutes with FT... ADD)

(Link: `https://en.wikipedia.org/wiki/Projection-slice_theorem`)

CTF:

Image model: $I = h * Pf + \epsilon$

FT: $\mathcal{F}I = \text{CTF} \cdot \mathcal{F}Pf + \mathcal{F}\epsilon$, where $\text{CTF} = \mathcal{F}h$

CTF affects frequencies differently: $\text{CTF}(f) = \sin(\chi(f)) e^{-Bf^2/4}$

(Link: Professor Singer's slides)

Kmeans, Kmeans++, EM:

Definition 3 (K-means objective)**.** *Given n points $x_1, \ldots, x_n \in \mathbb{R}^d$, want to find k cluster centers $\mu_1, \ldots, \mu_k$ s.t. the sum of cluster variances is minimized. I.e. if clusters are $S = \{S_1, \ldots, S_k\}$,*

$$\operatorname{argmin}_S \sum_{i=1}^{k} \sum_{x_j \in S_i} \|x_j - \mu_i\|^2 = \operatorname{argmin}_S \sum_{i=1}^{k} |S_i| \operatorname{Var}(S_i).$$

Lloyd's algorithm (Hard Kmeans): initialize centers. Then assign points to closest center. Then recompute center as average of assigned points. Rinse and repeat, until centers don't change.

Summary: Two step process: 1) estimate clusters, fixing centers; 2) estimate centers, fixing clusters. Continue until reach true centers.

1) ADD optimization prob...

2) ADD optimization prob...

(Initialization: Kmeans++... ADD)

EM algo (from TDS): It is used to maximize likelihood (to find MLE choice of parameters), approximately, when directly maximizing is intractable. Instead, introduce latent variables Δ to make it easier. Essentially maximize $\ell(\theta; X, \Delta)$, where Δ is the expected choice of latent variables.

Really, do in 2-step iterative algo: first find function $Q(\theta, \hat{\theta})$ to maximize by taking expectation to estimate Δ; then choose new parameters θ to maximize $Q(\theta, \hat{\theta})$.

(Q is "expected complete data log likelihood function".)

Summary: Two step process: 1) estimate latent, fixing parameters; 2) estimate parameters, fixing latent. Continue until reach true parameters.

Derivation: By Bayes, $P(X, \Delta|\theta) = P(\Delta|X, \theta)P(X|\theta)$. Take logs:

$$\log P(X|\theta) = \log P(X, \Delta|\theta) - \log P(\Delta|X, \theta),$$

i.e. log likelihood = complete data log likelihood − extra term (?), i.e.

$$\ell(\theta; X) = \ell(\theta|X, \theta) - \ell(\theta; \Delta|X).$$

LHS is difficult to maximize, but second term is easy to maximize... except don't know Δ! Replace with expected value, i.e. take $\mathbb{E}_{\Delta|X,\hat{\theta}}$, where $\hat{\theta}$ is current estimate of θ. Get

$$\ell(\theta; X) = \mathbb{E}_\Delta[\ell(\theta; X, \Delta)|X, \hat{\theta}] - \mathbb{E}_\Delta[\ell(\theta; \Delta|X)] = Q(\theta, \hat{\theta}) - R(\theta, \hat{\theta}).$$

To find update for θ, 1) solve for $Q(\theta, \hat{\theta})$, and 2) set $\hat{\theta} = \operatorname{argmax}_\theta Q(\theta, \hat{\theta})$.

Formal algo:

1. Fix $\hat{\theta}_0$.
2. For $j \geqslant 0$, until converge, repeat:
   (E-step) Compute $Q(\theta, \hat{\theta}_j) = \mathbb{E}_\Delta[\ell(\theta; X, \Delta)|X, \hat{\theta}_j]$
   (M-step) Set $\hat{\theta}_{j+1} = \operatorname{argmax}_\theta Q(\theta, \hat{\theta}_j)$

Why does it work? Because $\ell(\theta; X)$ always increases. Note $Q(\theta, \hat{\theta}_j)$ maximized at $\hat{\theta_{j+1}}$, while $R(\theta, \hat{\theta}_j)$ maximized at $\hat{\theta}_j$ (ADD proof using KL divergence/Jensen...). Thus the change in the log likelihood is

$$\ell(\hat{\theta}_{j+1}; X) - \ell(\hat{\theta}_j; X) = (Q(\theta, \hat{\theta}_{j+1}) - Q(\theta, \hat{\theta}_j)) - (R(\theta, \hat{\theta}_{j+1}) - R(\theta, \hat{\theta}_j)) = + - - = + > 0,$$

as desired.

Special cases of EM:

GMM: ADD... another place besides PPT?

Kmeans: ADD... see slide 34 for reduction from GMM...

The Kmeans problem is an example of ML because... MLE of GMM with hard thresholding... Lloyd's algorithm is a special case of EM algorithm to solve this problem...

Can also have Soft Kmeans... Special case of EM/GMM...

(Link: `https://www.quora.com/What-is-the-difference-between-soft-k-means-and-hard-k-means`)

(Link: `https://www.cs.cmu.edu/~epxing/Class/10715/lectures/EM.pdf`)

(Link: `https://towardsdatascience.com/expectation-maximization-explained-c82f5ed438e5`)

Kmeans as maximum likelihood/least squares problem? Why? Is it from the above hard thresholding version of GMM/EM?

Normal equations for centroid:

Original derivation (**might be flawed**):

Loss function $L(y) = \sum_i \left\| R_i \cdot \mathrm{CTF}_i y - \hat{I}_i \right\|^2$.

Define $A_i = R_i \cdot \mathrm{CTF}_i, b_i = \hat{I}_i$.

Then

$$L(y) = \sum_i \|A_i y - b_i\|^2 = \sum_i (A_i y - b_i)^T (A_i y - b_i) = \sum_i (y^T A_i^T - b_i^T)(A_i y - b_i)$$

$$= \sum_i (y^T A_i^T A_i y - y^T A_i^T b_i - b_i^T A_i y + b_i^T b_i).$$

Thus

$$\nabla_y L = \sum_i 2A_i^T A_i y - 2A_i^T b_i y = 0$$

$$\implies \hat{y} = \left(\sum_i A_i^T A_i\right)^{-1} \left(\sum_i A_i^T b_i\right),$$

like single variable calculus. (See formulae near end of `http://cs229.stanford.edu/section/cs229-linalg.pdf`)

(This comes up in different setting of LS. To solve $Ax = b$ approximately, you can find LS approximation by finding $\hat{x} = \operatorname{argmin}_x \|Ax - b\|^2$. This is solving a system of such matrix equations for y.)

Issue: need **complex** norm $\|z\|^2 = z\overline{z}$. Redo below:

Loss function $L(y) = \sum_i \left\| R_i \cdot \mathrm{CTF}_i y - \hat{I}_i \right\|^2$.

Define $A_i = R_i \cdot \mathrm{CTF}_i \in \mathbb{R}^{d^2 \times d^2}, b_i = \hat{I}_i \in \mathbb{C}^{d^2}$. We are trying to find $y \in \mathbb{C}^{d^2}$.

(As a $d \times d$ matrix, b_i and y are FT's of real signals, they are conjugate symmetric about their center entry. In 1d...)

Then

$$L(y) = \sum_i \|A_i y - b_i\|^2 = \sum_i (A_i y - b_i)^T (A_i \overline{y} - \overline{b}_i) = \sum_i (y^T A_i^T - b_i^T)(A_i \overline{y} - \overline{b}_i)$$
$$= \sum_i (y^T A_i^T A_i \overline{y} - y^T A_i^T \overline{b}_i - b_i^T A_i \overline{y} + b_i^T \overline{b}_i).$$

Using real variables:

Let $y = c + di$, $b_i = e_i + f_i i$.

$$y^T A_i^T A_i \overline{y} = c^T A_i^T A_i c - c^T A_i^T A_i d i + d^T A_i^T A_i c i + d^T A_i^T A_i d.$$

$$y^T A_i^T \overline{b_i} = c^T A_i^T e_i - c^T A_i^T f_i i + d^T A_i^T e_i i + d^T A_i^T f_i.$$

$$b_i^T A_i \overline{y} = e_i^T A_i c - e_i^T A_i d i + f_i^T A_i c i + f_i^T A_i d.$$

$$\begin{aligned} L(y) = \sum \Big(& c^T A_i^T A_i c - c^T A_i^T A_i d i + d^T A_i^T A_i c i + d^T A_i^T A_i d \\ & - (c^T A_i^T e_i - c^T A_i^T f_i i + d^T A_i^T e_i i + d^T A_i^T f_i) \\ & - (e_i^T A_i c - e_i^T A_i d i + f_i^T A_i c i + f_i^T A_i d) \Big) + \text{constant}. \end{aligned}$$

$$\nabla_c L(y) = \sum_i \left(2A_i^T A_i c - 2A_i^T e_i \right) = 0 \implies c = \left(\sum A_i^T A_i \right)^{-1} \left(\sum A_i^T e_i \right),$$

$$\nabla_d L(y) = \sum_i \left(2A_i^T A_i d - 2A_i^T f_i\right) = 0 \implies d = \left(\sum A_i^T A_i\right)^{-1} \left(\sum A_i^T f_i\right).$$

Thus, as above, as $\hat{y} = c + di$,

$$\hat{y} = \left(\sum A_i^T A_i\right)^{-1} \left(\sum A_i^T b_i\right)$$
$$\iff \hat{y} = \left(\sum \mathrm{CTF}_i^2\right)^{-1} \left(\sum \mathrm{CTF}_i \cdot R_i^{-1} b_i\right),$$

and our clean center estimate is

$$\hat{x} = \mathcal{F}^{-1}\hat{y} = ...?$$

(Sanity check: when no CTF/rotations needed, get $\hat{y} = \frac{1}{n}\sum_i b_i$, the average.)

Links about Wiener filter:

`https://www.probabilitycourse.com/chapter10/10_2_1_power_spectral_density.php` (Power spectral density + autocorrelation)

`https://www.youtube.com/watch?v=L71HfwPlkuM` (Power spectrum)

`https://www.youtube.com/watch?v=GE3_4acUrO4` (DIP lecture 17)

`http://sdeuoc.ac.in/sites/default/files/sde_videos/Digital%20Image%20Processing%203rd%20ed.%20-%20R.%20Gonzalez%2C%20R.%20Woods-ilovepdf-compressed.pdf` (DIP textbook)

Kernel function

Laplacians on domains (disc, rectangle, spherical harmonics), Laplacian eigenfunctions [affinity matrix = block matrix?]

Laplace–Beltrami operator (Boumal optimization on manifolds book)

Spectral clustering (see gdoc for next few)

Cheeger's inequality

Diffusion maps

VDM:

(Singer, Zhao, Rotationally invariant...)

(Link: `https://www.youtube.com/watch?v=bXKyW2fS5bs`)

Imagine a graph of nodes, where node i corresponds to projection P_i. We initially draw edges s.t. we connect i, j if they k nearest neighbors wrt d_{RID}, the rotationally invariant distance. However, due to noise, we will have a few outlier edges that connect very different viewing directions.

To remove the outlier edges, we define a new metric/affinity between i, j. It is based on the consistency of paths between i, j.

Let α_{ij} be the in-plane rotation that takes i to j. Define

$$H_{ij} = \begin{cases} e^{i\alpha_{ij}} & \text{if } i \sim j, \\ 0 & \text{else.} \end{cases}$$

Lemma 4. *The matrix H is Hermitian.*

Proof. Using $\alpha_{ij} + \alpha_{ji} \equiv 0 \pmod{2\pi}$, we have

$$\begin{aligned} (H^*)_{ij} &= (\overline{H})^T_{ij} = \overline{H}_{ji} \\ &= \begin{cases} e^{-i\alpha_{ji}} & \text{if } j \sim i, \\ 0 & \text{else,} \end{cases} \\ &= \begin{cases} e^{i\alpha_{ij}} & \text{if } i \sim j, \\ 0 & \text{else,} \end{cases} \\ &= H_{ij}, \end{aligned}$$

as desired. □

Note H has $O(nk)$ elements and is therefore sparse.

Define the diagonal degree matrix D s.t. $D_{ii} = \deg(i) = \sum_j |H_{ij}|$. Define $S = D^{-1}H$. As $S = D^{-\frac{1}{2}}(D^{-\frac{1}{2}}HD^{-\frac{1}{2}})D^{\frac{1}{2}}$, S is similar to $\tilde{S} = D^{-\frac{1}{2}}HD^{-\frac{1}{2}}$. Note $\tilde{S}$ is hermitian, as $\tilde{S}^* = D^{-\frac{1}{2}}H^*D^{-\frac{1}{2}} = \tilde{S}$.

Given $t \in \mathbb{N}$, we define the affinity between i, j as $\left|\tilde{S}^{2t}(i,j)\right|^2$. This measures the agreement between in-plane rotations along all paths of length $2t$ from i to j, as each entry of S represents an in-plane rotation.

To compute the affinity, need the spectral decomposition of $\tilde{S}$. As $\tilde{S}$ hermitian, the Spectral Theorem implies $\tilde{S} = V\Lambda V^*$, where V is a unitary matrix of orthonormal eigenvectors and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ (sorted in decreasing order of magnitude). Expanding the matrix multiplication and choosing the i, j entry, we find $\tilde{S}(i,j) = \sum_{l=1}^n \lambda_l v_l(i)\overline{v_l(j)}$. As $\tilde{S}^{2t} = V\Lambda^{2t}V^*$, we also get $\tilde{S}^{2t}(i,j) = \sum_{l=1}^n \lambda_l^{2t} v_l(i)\overline{v_l(j)}$. Thus the affinity is

$$\left|\tilde{S}^{2t}(i,j)\right|^2 = \left|\sum_{l=1}^n \lambda_l^{2t} v_l(i)\overline{v_l(j)}\right|^2.$$

This can be rewritten in a different way, which explains the term "vector diffusion **map**". Define the "vector diffusion map" V_t that sends a node to a vector in the Hilbert space (complete inner product space) $\mathbb{C}^{n^2}$ via $V_t : i \mapsto ((\lambda_l\lambda_r)^t v_l(i)\overline{v_r(i)})_{l,r\in[n]}$. Then interestingly, if we compute $\langle V_t(i), V_t(j)\rangle$ (where we use the standard inner product $\langle u, v\rangle = \sum u_i\overline{v_i}$), we find

$$\langle V_t(i), V_t(j)\rangle = \sum (\lambda_l\lambda_r)^{2t} v_l(i)\overline{v_r(i)}\,\overline{v_l(i)\overline{v_r(i)}}$$

$$
\begin{aligned}
&= \sum \lambda_l^{2t} v_l(i)\overline{v_l(j)} \cdot \lambda_r^{2t}\overline{v_r(i)}v_r(j) \\
&= \left(\sum \lambda_l^{2t} v_l(i)\overline{v_l(j)}\right)\overline{\left(\sum \lambda_l^{2t} v_r(i)\overline{v_r(j)}\right)} \\
&= \tilde{S}^{2t}(i,j)\overline{\tilde{S}^{2t}(i,j)} = \left|\tilde{S}^{2t}(i,j)\right|^2,
\end{aligned}
$$

the affinity! So the complicated affinity between i, j is just the inner product of the mapped vectors $V_t(i), V_t(j)$.

Computationally, we can approximate the affinity by only using the m eigenvectors corresponding to the largest $|\lambda_i|$.

We use this metric to improve our original kNN.

We also use this metric to estimate the in-plane rotations within each class. ADD...

ADD... outer product derivation of spectral decomposition.

Singer–Zhao: rotationally invariant image representation:

ADD...

[Add notes on top of stack]

MAT 586 slides:

PCA and KPCA:

ADD...

Proof of Laplacian convergence:

Proposition 5 (At the center of a ball of radius r in $\mathbb{R}^p$)**.** *Suppose our points $\{x_i\} \subset \mathbb{R}^p$ are uniformly distributed in the ball $B(x_0, r)$ of radius r. Let K denote the kernel function. Then the discrete Laplacian at x_0 for $\{x_0, \ldots, x_n\}$ is*

$$\frac{1}{n}\sum_{i=1}^{n}(f(x_i)-f(x_0))\,K\left(\frac{\|x_i-x_0\|}{\epsilon}\right) \to \int_{B(x_0,r)}(f(x)-f(x_0))\,K\left(\frac{\|x-x_0\|}{\epsilon}\right)\,dx.$$

Moreover, the limiting integral reduces to

$$\left(\frac{1}{2}\int_{B(0,r)} x_1^2 K\left(\frac{\|x\|}{\epsilon}\right)\,dx\right)\Delta f(x_0).$$

Proof. Taylor expanding f around x_0, we find

$$f(x) \approx f(x_0) + \nabla f(x_0)\cdot(x-x_0) + \frac{1}{2}(x-x_0)^T H(x_0)(x-x_0),$$

where H is the Hessian matrix of f. As

$$\nabla f(x_0)\cdot(x-x_0)K\left(\frac{\|x-x_0\|}{\epsilon}\right)$$

is an odd function of $(x - x_0)$ on the radially symmetric region, our integral reduces to

$$\begin{aligned}
&\int_{B(x_0,r)} \frac{1}{2}(x-x_0)^T H(x_0)(x-x_0)K\left(\frac{\|x-x_0\|}{\epsilon}\right)\,dx \\
&\quad= \int_{B(0,r)} \frac{1}{2}x^T H(x_0)xK\left(\frac{\|x\|}{\epsilon}\right)\,dx \\
&\quad= \sum_{1\leqslant i,j\leqslant p}\int_{B(0,r)} \frac{1}{2}\partial_{ij}f(x_0)x_ix_jK\left(\frac{\|x\|}{\epsilon}\right)\,dx \\
&\quad= \sum_{i=1}^{p}\int_{B(0,r)} \frac{1}{2}\partial_{ii}f(x_0)x_i^2K\left(\frac{\|x\|}{\epsilon}\right)\,dx \\
&\quad= \left(\frac{1}{2}\int_{B(0,r)} x_1^2K\left(\frac{\|x\|}{\epsilon}\right)\,dx\right)\Delta f(x_0),
\end{aligned}$$

as desired. □

Lemma 6 (Radial symmetry of the second moment)**.** *For a radial kernel on the ball,*

$$\int_{B(0,r)} x_i^2K\left(\frac{\|x\|}{\epsilon}\right)\,dx = \int_{B(0,r)} x_j^2K\left(\frac{\|x\|}{\epsilon}\right)\,dx.$$

Proof. Consider the change of variables given by $x_k = u_k$ if $k \neq i, j$, $x_i = u_j$, and $x_j = u_i$. This is an isometry, so $\|x\| = \|u\|$ and the Jacobian determinant is $\frac{\partial x}{\partial u} = -1$. It follows that

$$\int_{B(0,r)} x_i^2K\left(\frac{\|x\|}{\epsilon}\right)\,dx = \int_{B(0,r)} u_j^2K\left(\frac{\|u\|}{\epsilon}\right)\,du,$$

as desired. □

Proposition 7 (At the origin on a face of a rectangular box $R \subset \mathbb{R}^p$)**.** *As above, the empirical Laplacian converges to*

$$\int_R (f(x) - f(0))\, K\left(\frac{\|x\|}{\epsilon}\right) dx.$$

If the origin is on the center of the face, then as $r \to 0$, only the first-order term remains, and the Laplacian becomes

$$\left(\int_R x_p K\left(\frac{\|x\|}{\epsilon}\right) dx\right) \partial_p f(0).$$

Proof. Taylor expanding f around 0, we find

$$f(x) \approx f(0) + \nabla f(0) \cdot x + \frac{1}{2} x^T H(0) x.$$

Our integral reduces to

$$\begin{aligned}
&\int_R \left(\nabla f(0) \cdot x + \frac{1}{2} x^T H(0) x\right) K\left(\frac{\|x\|}{\epsilon}\right) dx \\
&\quad = \sum_{i=1}^{p} \left(\int_R x_i K\left(\frac{\|x\|}{\epsilon}\right) dx\right) \partial_i f(0) \\
&\qquad + \sum_{1 \leqslant i,j \leqslant p} \left(\frac{1}{2} \int_R x_i x_j K\left(\frac{\|x\|}{\epsilon}\right) dx\right) \partial_{ij} f(0),
\end{aligned}$$

a sum of a scaled divergence, a scaled Laplacian, and scaled mixed second order terms.

If the origin is on the center of the face, let x_p denote the normal direction and $x_1, \ldots, x_{p-1}$ denote the directions in the plane of the face. There is symmetry with respect to x_i for all $i < p$, and all terms with an odd exponent on at least one x_i cancel out. Our Laplacian reduces to

$$\left(\int_R x_p K\left(\frac{\|x\|}{\epsilon}\right) dx\right) \partial_p f(0) + \sum_{i=1}^{p} \left(\frac{1}{2} \int_R x_i^2 K\left(\frac{\|x\|}{\epsilon}\right) dx\right) \partial_{ii} f(0).$$

If r is the largest side-length of R, $x_i^2 = O(r^2)$ while $x_i = O(r)$. As $r \to 0$, only the first order term remains, and our Laplacian becomes

$$\left(\int_R x_p K\left(\frac{\|x\|}{\epsilon}\right) dx\right) \partial_p f(0).$$

□

...

Proposition 8 (At the center of a ball of radius r in $\mathbb{R}^p$, but with a non-radially symmetric kernel)**.** *Suppose our kernel is given by $K(x)$, i.e. not radially symmetric. As above, the empirical Laplacian converges to*

$$\int_{B(0,r)} (f(x) - f(0))\, K\left(\frac{x}{\epsilon}\right) dx.$$

The corresponding expansion is a sum of a scaled divergence, a scaled Laplacian, and scaled mixed second order terms.

Proof. Taylor expanding f around 0, we find

$$f(x) \approx f(0) + \nabla f(0) \cdot x + \frac{1}{2} x^T H(0) x.$$

Our integral becomes

$$\begin{aligned}
&\int_{B(0,r)} \left(\nabla f(0) \cdot x + \frac{1}{2} x^T H(0) x \right) K\left(\frac{x}{\epsilon}\right) \, dx \\
&\qquad = \sum_{i=1}^{p} \left(\int_{B(0,r)} x_i K\left(\frac{x}{\epsilon}\right) \, dx \right) \partial_i f(0) \\
&\qquad\quad + \sum_{1 \leqslant i,j \leqslant p} \left(\frac{1}{2} \int_{B(0,r)} x_i x_j K\left(\frac{x}{\epsilon}\right) \, dx \right) \partial_{ij} f(0),
\end{aligned}$$

a sum of a scaled divergence, a scaled Laplacian, and scaled mixed second order terms. □